FORMATIVE ASSIGNMENT

Education Without Barriers

- **Description of the concept of their Mission**

  My mission is to improve the education sector using language. I want to make education more accessible and more understandable by providing a mother-tongue version of an already existing English version of the national curriculum.

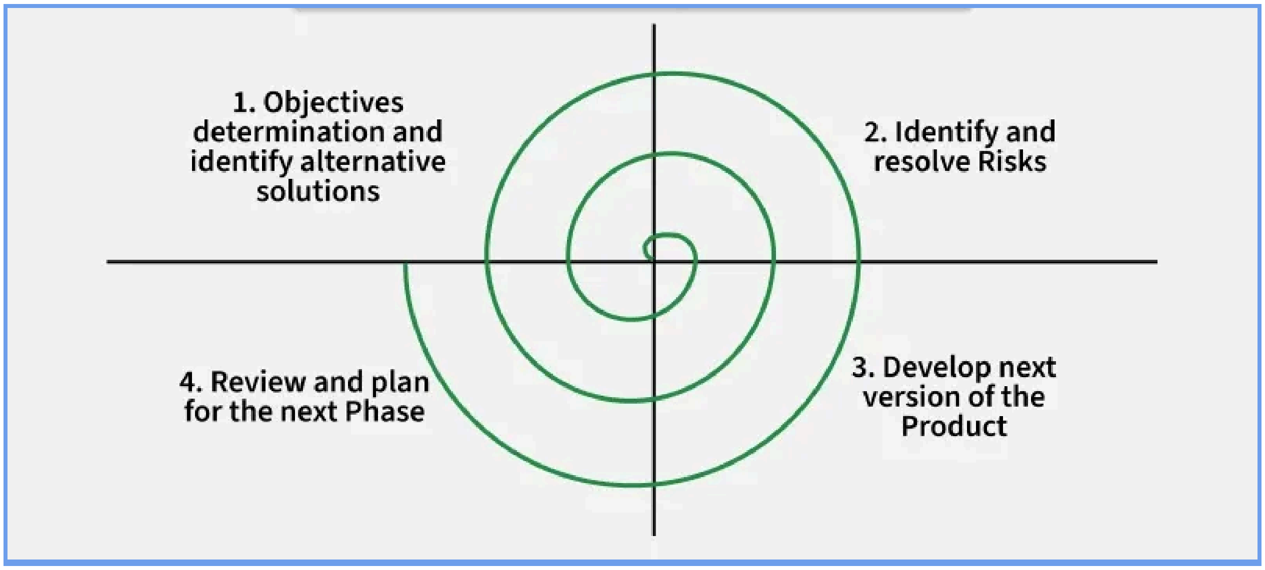
- **Problem statement:**

  You show up for your first day at school, and the teacher starts talking to you in a language you don't understand. You're now expected to learn to read and write in this language. That's exactly the scenario they face when starting school, and the outcome is disastrous. UNESCO estimates that 40% of school-aged children don't have access to education in a language that they understand, and it estimates that if all students had basic literacy skills, 171 million people could escape extreme poverty. Despite years of schooling, many students end up essentially illiterate because of language barriers(Concern Worldwide, n.d.).

  Students learning a second language often struggle to express themselves if they don't have a full command of that language, notes John Schumann of UCLA's Department of Applied Linguistics. This can lead to emotional stress and affect their ability to learn(Borgen Magazine, n.d.).

  The main reason provided for providing education in a foreign language is that it exposes the student to global opportunities, but that comes at a cost. Making education materials incomprehensible to 40% of the students.
  Some African countries like Zimbabwe, Madagascar, Rwanda, and Zambia have already started to shift towards their own mother tongue, and since 2014, Zambian authorities have noted higher test scores in math, reading, and writing in primary schools(Borgen Magazine, n.d.).

- **Software development model**



Spiral Model: We have 4 quadrants where we circle among them multiple times from 1. Objective determination to identify and resolve risks to Development, to customer feedback, and the same cycle is repeated over and over till the customer is satisfied. The Spiral Model emphasises iterative development and continuous risk assessment, making it suitable for projects where requirements may evolve. By applying the spiral model, we ensure that our customer satisfaction.

1. **Step 1:** Here, we work on identifying stakeholders, determining curriculum requirements, identifying languages supported, and defining scope while delivering requirements specifications.
2. **Step 2:** Here, we address the translation inaccuracies, copyright concerns, and low translator participation, while also mitigating the experts' reviews and delivering the risk management report.
3. **Step 3:** Development involves user authentication, the translation module, a book repository, a review system, and search functionality, and at the end we deliver a working prototype. Here we build the Build app, database, and APIs.
4. **Step 4:** We test the product with teachers, students, and translators and get feedback on what to improve. Then, depending on the feedback, we go back to the first step.

- **The hypothesis of their solution**

  If many curriculum books are translated into Kinyarwanda and made available through the Education Without Barriers platform, then students participating in a six-month pilot programme involving 300 primary and lower secondary learners in Rwanda will demonstrate at least a 20% improvement in reading comprehension scores, a 15% increase in classroom participation, and a 25% increase in access to educational materials compared to baseline measurements.

- **References**

Concern Worldwide. (n.d.). *Language barriers in the classroom*. https://concernusa.org/news/language-barriers-in-classroom/
Borgen Magazine. (n.d.). *Mother tongue education*. https://www.borgenmagazine.com/mother-tongue-education/

Software Requirements Specification (SRS) Template

Education Without Barriers

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05/18/2026

Revision History

Name	Date	Reason For Changes	Version
Jean Nepo Munezero	18/05/2026	Creation of the original Version	Version 1.0.1

1. Introduction

1.1 Purpose

The product specified in this Software Requirements Specification (SRS) is an educational mobile and web application called Education Without Barriers, Version 1.0.

The application is designed to help educational stakeholders, especially students, teachers, and parents, access curriculum-based educational materials in their mother tongue. The system provides translated versions of existing educational books, which are commonly available only in English, in order to improve understanding and learning outcomes.

This SRS covers the core functionalities of the system:

- User registration and authentication
- Browsing and searching for educational books
- Ability to translate books
- Downloading translated learning materials
- Switching the application language to the user's preferred mother tongue
- Viewing the most popular books
- Reviewing or rating books

1.2 Document Conventions

FR: Functional Requirements
NFR: Non-Functional Requirements
SRS: Software Requirements Specification
Pr2: Project proposal

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for different categories of readers involved in the development and use of the system. These include **developers, project managers, testers, system administrators, documentation writers,** and **end users.**

- **Developers** will use this document to understand the technical requirements of the software, functional and non-functional requirements of the system, the technologies involved and their requirements, hardware requirements, and the expected behavior of the application.
- **Project managers** will use the document to understand the project scope, objectives, requirements, and system constraints for planning and monitoring purposes.
- **Testers** will refer to the requirements and system specifications to design software testing activities.
- **Documentation writers** will use this Pr2 document to design user manuals and technical documentation.
- **End users and stakeholders** will use the overview sections to understand the purpose, features, and benefits of the system.

This document is organized into sections that describe the overall system requirements, functional requirements, non-functional requirements, external interface requirements, and system features.

It is intended to provide a thorough understanding of the software platform, covering the technical aspects of the software and the non-technical aspects of the final product. It is intended to answer multiple questions regarding the requirements and performance of the software on multiple devices. The provided order has the external software requirements covered in section **3. External Interface Requirements.**

1.4 Product Scope

Education Without Barriers is a mobile and web-based educational platform that provides curriculum-based learning materials translated into local languages such as Kinyarwanda. The system is designed to help students and other educational stakeholders better understand educational content by reducing language barriers.

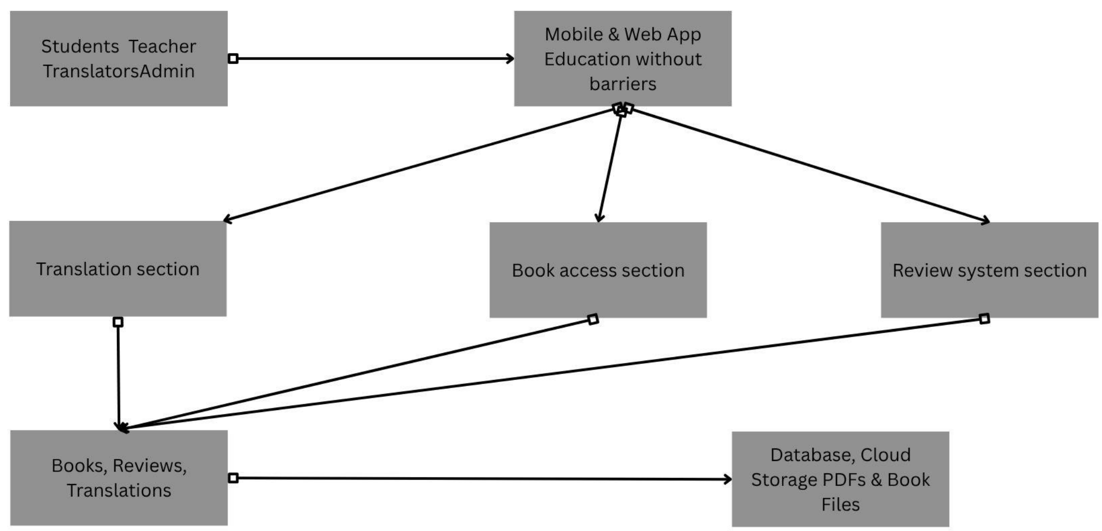
The platform allows users to search, download, review, and translate educational books in their mother tongue. Its main goal is to improve accessibility, understanding, and appreciation of education while supporting local language development and inclusive learning.

2. Overall Description

2.1 Product Perspective

This platform makes it possible to get access to educational materials in one's mother tongue in order to make education more appealing in response to the challenges students face when learning from educational content written mainly in English. It serves as a digital library and an open-source collaboration platform.

It includes components such as a user interface, translation module, book management system, review system, and database for storing educational materials and user information.



2.2 Product Functions

- Users can download a PDF version of a book
- Freelancers can submit translated versions of text.
- Users can customize the nomenclature of our system
- Users can review a book
- Browsing and searching for educational books
- Viewing the most popular and highly rated books
- Admin management of users, books, and translations

2.3 User Classes and Characteristics

- **Students:** Students are the primary users of the platform; they will read books from our platform and be able to download them for free. They can review, rate, and translate some materials.
- **Teachers:** Teachers will have the ability to get alternative options to their curriculum books. They can download already existing books for personal use, review and rate them, and perform text translation.
- **Translators:** They can translate given chunks of text to be reviewed.
- **Parents:** They may use the platform to access learning materials and support students in their studies.
- **Government entity:** The government is a higher authority that designs the national learning curriculum in English, and it will use our platform to assess our credibility and authenticity.

- **Educational stakeholders and Language enthusiasts:** Since our platform is open-source, any users will be able to get content out of it for free. For example, we will welcome language enthusiasts who want to use it to train their models.

2.4 Operating Environment

Hardware Platform

- Smartphones and tablets
- Desktop and laptop computers
- Cloud or local servers for data storage and management

Operating Systems

- Android 10 or above
- Windows 10 or above
- Linux and macOS systems
- Modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari

Software Requirements

- Stable internet connection for downloading and uploading books
- Database management system for storing user and book information
- Web server and application server technologies
- Translation and file management modules

The software will coexist peacefully with other applications installed on the user's device without affecting system performance.

2.5 Design and Implementation Constraints

The development of the Education Without Barriers platform will be subject to several constraints and limitations.

- The system will support multiple languages, especially Kinyarwanda, which may require additional translation.
- The application will operate on devices with limited memory and processing power, especially low-end Android smartphones commonly used by students.
- A stable internet connection is required for downloading and uploading books, although some features may support offline access.

- The platform will use secure authentication and data protection measures to protect user information and uploaded content.
- The system will be compatible with modern web browsers and Android operating systems.
- The project may rely on cloud storage and database technologies for managing books, translations, and user data.
- Copyright and educational content regulations will be respected when translating and distributing curriculum materials.
- The software should follow standard programming and interface design practices to ensure maintainability and usability.

2.6 User Documentation

The Education Without Barriers platform will provide several forms of user documentation to assist users in understanding and using the system effectively.

The documentation components will include:

- User manual
- Installation and setup guide
- Online help section
- Frequently Asked Questions (FAQ)
- Tutorials for downloading, reading, reviewing, and translating books
- Administrator's guide for managing users and content

The documentation will be delivered in digital formats such as PDF documents, in-app help pages, and web-based tutorials.

2.7 Assumptions and Dependencies

Assumptions

- Users will have access to smartphones, computers, or tablets with internet connectivity.
- Translators will be reliable and professional.
- A large number of Educational books and curriculum materials will be legally available for translation and distribution.
- Users contributing translations will have sufficient language knowledge and translation skills.
- The system will mainly be used in Rwanda during the initial release, with Kinyarwanda as the primary local language.
- Users will have basic digital literacy skills to operate the application.

Dependencies

- The platform relies on cloud storage and database services to store books, translations, and user information.
- The system may rely on third-party libraries, translation tools, and authentication services.
- The application depends on stable internet services for uploading and downloading educational materials.
- Future system improvements may depend on integration with educational institutions and online learning platforms.
- The project depends on the continued availability of supported operating systems, web browsers, and mobile technologies.

3. External Interface Requirements

3.1 User Interfaces

The interface will use simple layouts, readable fonts, clear navigation menus, and standard buttons such as Login, Register, Download, Translate, Search, Review, and Help. Users will be able to switch the system language to their preferred mother tongue, such as Kinyarwanda.

The main interface components will include:

- Login and registration screens
- Home page and dashboard
- Book browsing and search sections
- Book reading and downloading pages
- Translation interface for translators
- Review and rating interface
- Administrator management panel

The system should display clear validation and error messages when incorrect information is entered. The user interface should remain responsive and compatible across smartphones, tablets, and desktop computers. Some of the status notifications with their respective literal meanings:

- Red notification: "Please use a valid phone number."
- Red notification: "Incorrect username or password."
- Red notification: "Internet connection lost."
- Red notification: "This field cannot be empty."
- Red notification: "Book upload failed. Please try again."
- Green notification: "Login successful."
- Green notification: "Book downloaded successfully."
- Green notification: "Translation submitted successfully."
- Green notification: "Review posted successfully."
- Yellow notification: "Your session is about to expire."
- Yellow notification: "Download paused due to weak internet connection."
- Yellow notification: "This translation is awaiting approval."

Detailed screen designs and layouts will be provided in a separate User Interface Design Specification document.

3.2 Hardware Interfaces

The Education Without Barriers platform will interact with various hardware components through mobile and web technologies. The system is designed to operate on smartphones, tablets, laptops, and desktop computers.

Supported Devices

- Android smartphones and tablets
- Desktop and laptop computers
- Cloud and database servers

Hardware Interaction

The software will interact with:

- Device storage for saving downloaded books
- Display screens for graphical user interfaces
- Touchscreens, keyboards, and mouse devices for user input
- Network hardware for internet communication

Communication Protocols

The system will use standard communication protocols such as HTTP/HTTPS and TCP/IP for secure data transfer between user devices and servers.

The application should perform efficiently on low-end devices with limited hardware resources.

3.3 Software Interfaces

The platform will interface with several software components and services required for system operation.

Software Components

- Android operating system
- Web browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge
- Cloud database management systems such as PostgreSQL or MySQL
- File storage services for educational materials
- Authentication and login services
- Translation support tools and libraries

Libraries and Frameworks

- React or Next.js for the web frontend
- React Native or Flutter for the mobile application
- Node.js and Express.js for backend services
- Tailwind CSS or Bootstrap for interface styling
- Axios or Fetch API for API communication

Tools and Platforms

- Git and GitHub for version control
- Visual Studio Code for software development
- Postman for API testing
- Figma for interface design
- Firebase or Supabase for backend support and authentication

Databases

- PostgreSQL
- MySQL
- Firebase Firestore
- Supabase Database

Data Sharing

The system will exchange data such as:

- User registration and login information
- Educational books and translated content
- User reviews and ratings
- Download and upload requests

Communication Between Components

The application will communicate with servers and databases through APIs and secure web services. Data exchanged between software components should be encrypted and securely stored using secure communication methods.

3.4 Communications Interfaces

The Education Without Barriers platform requires internet communication for user authentication, downloading and uploading books, submitting translations, sending reviews, and accessing cloud databases.

Communication Standards

The system will use:

- HTTPS for secure web communication
- WebSocket for real-time notifications and updates
- MQTT for lightweight messaging where needed
- RESTful APIs for client-server communication
- TCP/IP for internet connectivity
- Email services for account verification and notifications

Security and Data Format

- User data will be encrypted during transmission.
- Secure authentication and access control mechanisms will be implemented.
- Data exchanged between the application and server will use JSON formatting for efficient communication.

4. Requirement Specification

STAKEHOLDER REQUIREMENTS SPECIFICATION

Functional Requirements

Req ID	Requirements	Description
FR 1	User Registration	Allow users to create an account in the system using an email or a phone number
FR 2	User Login	Allow registered users to log into the system securely
FR 3	Language Selection	Allow users to select and switch the system language (e.g., English, Kinyarwanda)
FR 4	Book Management	Provide access to educational books available in the system
FR 4.1	Browse Books	Allow users to view available curriculum books
FR 4.2	Search Books	Allow users to search for books by title, subject, or category
FR 4.3	Download Books	Allow users to download books for offline reading
FR 5	Book Reading	Allow users to open and read books within the application
FR 6	Translation Functionality	Allow users to translate selected books into their mother tongue
FR 6.1	Submit Translation	Allow users to submit translated book content
FR 6.2	Review Translation	Allow admins or teachers to review and approve translations
FR 7	Review System	Allow users to rate and review books
FR 8	Popular Books Display	Display the most downloaded or highest-rated books on the home screen
FR 9	Notifications	Send system notifications such as success, error, and update messages
FR 10	Admin Management	Allow administrators to manage users, books, and translations

5. Other Nonfunctional Requirements

5.1 Example of Non-Functional Requirements

Requirement Type	Req ID	Description
Security	NFR 1	The system will authenticate users securely using mail/phone login and encrypted credentials
Security	NFR 2	User data (login details, reviews, translations) will be encrypted during storage and transmission
Performance	NFR 3	The system will support multiple concurrent users (at least 100 users logged in simultaneously without performance degradation)
Usability	NFR 4	The system will be simple and user-friendly, with clear navigation suitable for students and teachers
Usability	NFR 5	The system will support multiple languages, including English and Kinyarwanda
Reliability	NFR 6	The system will ensure high availability with minimal downtime during normal operation
Auditability	NFR 7	The system will keep logs of user actions such as login, downloads, translations, and reviews
Cross-Browser Support	NFR 8	The web application will work on modern browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge
Compatibility	NFR 9	The system will run on Android devices (Android 10+) and desktop platforms (Windows, macOS, Linux)
Portability	NFR 10	The mobile application will be installable via standard app distribution methods and will function on low-end devices
Scalability	NFR 11	The system will be scalable to support future expansion to more countries and languages

Performance Requirements

The system will meet the following performance expectations under normal operating conditions:

- The application will load the home page within **3 seconds** on a standard internet connection.
- Book search results will be returned within **2 seconds**.
- The system will support at least **100 concurrent users** without performance degradation.
- Book download operations will start within **5 seconds** after the user request (depending on network speed).
- Translation submission requests will be processed within **10 seconds** under normal server load.
- The system will remain responsive on low-end Android devices with limited memory.

Safety Requirements

The system will ensure that no harm or loss results from its usage:

- The application will prevent accidental deletion of educational content by requiring confirmation before critical actions.
- Downloaded files will be scanned or validated to avoid corrupted or harmful data.
- The system will not display misleading or inappropriate educational content.
- User-generated translations will go through approval before being published to avoid misinformation.
- The system will ensure safe usage for minors (students), including protection from harmful or unauthorized content.

External Considerations:
The system will comply with general digital safety and child protection guidelines applicable to educational platforms.

Security Requirements

- Users will authenticate using secure login (email/phone and password).
- Translation permission will be sought from an authorised body, such as the book owners.
- Passwords will be stored using encryption or hashing techniques.
- All communication between client and server will use **HTTPS encryption**.
- Role-based access control will be implemented (student, teacher, translator, admin).
- Users should only access features allowed by their role.
- The system will protect against unauthorised access, data leaks, and injection attacks.
- Sensitive user data will not be publicly accessible.

Software Quality Attributes

The system will meet the following quality standards:

- **Usability:** The system will be easy to use for students with basic digital skills.
- **Reliability:** The platform should have at least **99% uptime** during school periods.
- **Performance:** Fast response time for search, reading, and download operations.
- **Scalability:** The system will support future expansion to more countries and languages.
- **Maintainability:** The codebase will be modular and easy to update.
- **Portability:** The system will run on Android, web browsers, and low-end devices.

- **Interoperability:** The system will integrate with cloud storage and translation APIs if needed.
- **Availability:** The system should remain accessible even under moderate network instability.

Business Rules

The system will operate under the following rules:

- Only registered users can download books or submit translations.
- Only verified translators or approved users can publish translations.
- Administrators have full control over books, users, and translations.
- Students can read and review books but cannot modify official content.
- All translations will be reviewed before being made public.
- Educational content will follow curriculum guidelines before publication.
- Users will respect the intellectual property rights of original educational materials.

6. Appendix

Requirements
The system will use a cloud or relational database (e.g., Firebase, PostgreSQL, or MySQL) to store users, books, translations, and reviews. The database will support fast access, secure storage, and regular backups.

Internationalization Requirements
The system will support multiple languages, starting with English and Kinyarwanda. Users will be able to switch languages, and all system content will be translatable.
Legal Requirements

The system will respect copyright laws for educational materials. Only authorised content may be uploaded or translated, and user data will be protected according to privacy regulations.

Reuse Objectives

The system should be modular, so components like authentication, book management, and translation can be reused or extended in future versions.

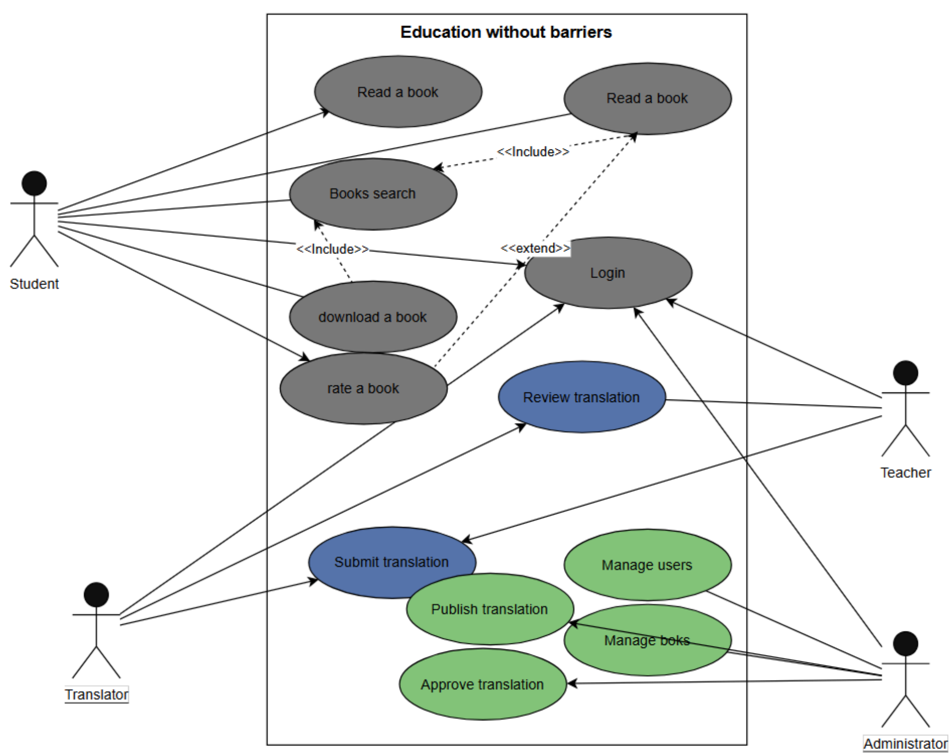
Appendix A: Glossary

- **SRS:** Document describing system requirements.

- **User:** Any system user (student, teacher, admin, translator).
- **Admin:** User with full system control.
- **Translator:** A user who translates books.
- **API:** Software interface for communication between systems.
- **Database:** A system for storing data.
- **Authentication:** Verifying user identity during login.
- **Encryption:** Protecting data from unauthorised access.
- **UI:** User interface (what users see and interact with).

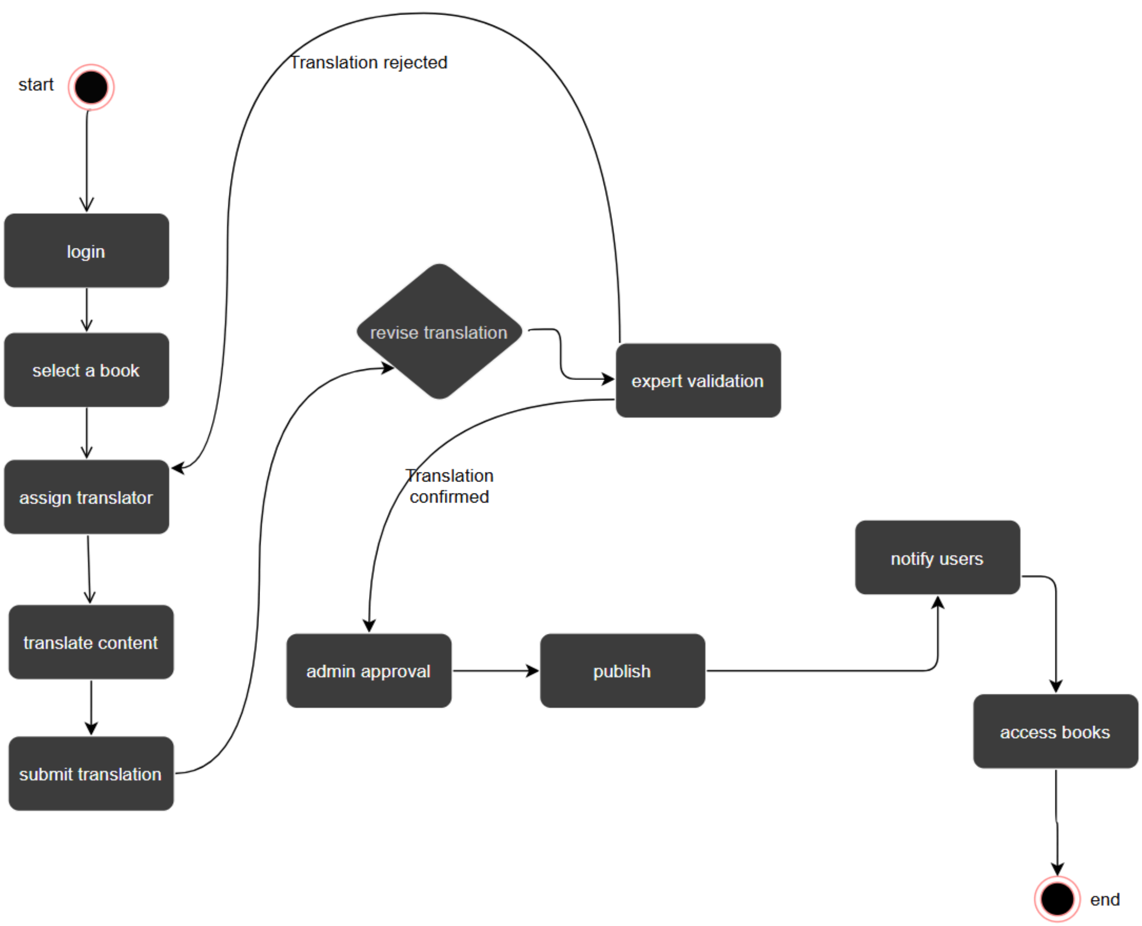
Appendix B: Analysis Models

B.1 Use Case Diagram



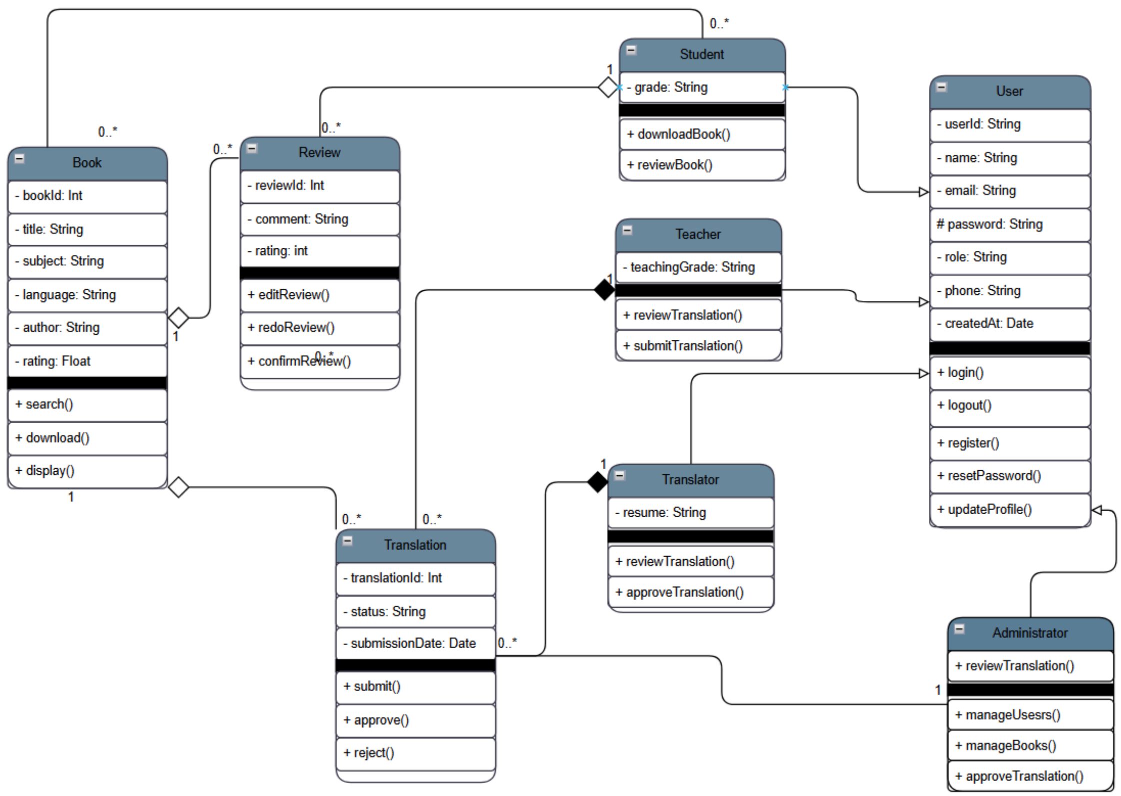
It demonstrates how Teachers, students, and translators will interact with the Education Without Barriers platform. It identifies the services available to each user. Students can browse, read, download, review, and rate books. Translators can submit translated content, while teachers and administrators review and approve translations before publication. Administrators manage books, translations, and user accounts.

B.2 Activity diagram



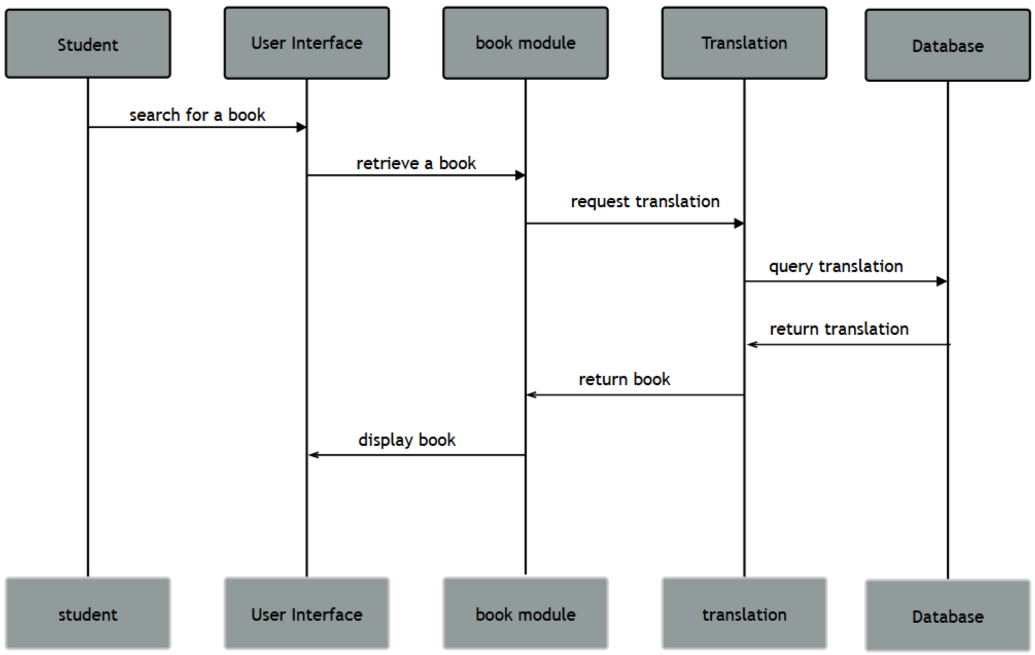
It visually models the sequential workflow or step-by-step logic of the Education Without Barriers platform. represents the workflow used to translate educational materials. It shows the sequence of activities performed from selecting a book to publishing the approved translation.

B.3 Class Diagram



– The Class Diagram illustrates the static structure of the Education Without Barriers platform. It shows the classes within the system, their attributes, methods, inheritance relationships, and associations.

B.4 Sequence Diagram



This sequence diagram illustrates the interaction between the student, the user interface, system modules, and the database when searching for and accessing a translated educational resource. It shows the flow of requests and responses required to retrieve and display the selected book.